Shore Thing Accessible Game Design

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INTRODUCTION

There is a growing need for accessible and engaging climate change education that connects to local environments, making the issue relevant and actionable for communities. This approach should emphasize local ecosystems and foster a message of acceptance, ensuring that everyone feels empowered to contribute to positive change. By highlighting shared responsibility and environmental stewardship, we can inspire individuals to take meaningful action in addressing climate change.

SHORE THING is a collaboration between five team members who participated in the a2RU Student Creative Summit in March 2024.

This project began as a response to the 2024 creative summit theme, “Play”.

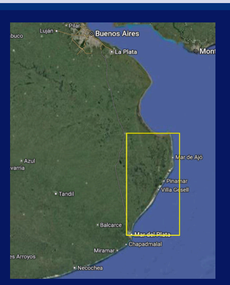
LOCATIONS



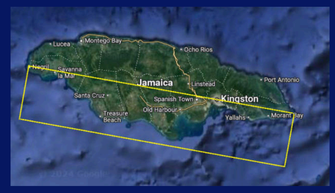
Rhode Island, USA
(sea level rising ~2.56 mm/yr)
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Mar Del Plata, Argentina
(sea level rising ~1.28 mm/yr)
Does not experience hurricanes; is not subsiding. Stabilized by saltmarshes.



Georgia, USA
(sea level rising 3.61 mm/yr)
Experiences subsidence, hurricanes, and tropical storms. Stabilized by saltmarshes and cold water corals.



South Shore, Jamaica
(sea level rising 3.3 mm/yr)
Experiences hurricanes and tropical storms. Is not subsiding. Stabilized by mangroves and warm water corals.



BACKGROUND

Coastal barriers are essential for protecting the land from storms and erosion while providing crucial habitats for wildlife. They support biodiversity and offer resources for humans, such as fisheries and recreational spaces. Protecting these ecosystems is vital for both environmental health and the well-being of coastal communities.

OF WORLD COASTLINES

20%

BARRIES BEACHES AND ISLANDS

Factors that contribute to sea level and coastal changes

Thermal expansion of seawater: the total volume of the ocean expands as it heats

Glacier melting: the melting of land ice adds water to the ocean. Sea ice melting does not impact global sea levels.

Mountain glaciers (1% of land ice)
Continental ice caps (99% of land ice)

Land Subsidence:
post-glacial isostatic adjustment is causing some coastlines to sink relative to their current level while others rise.

Storm Events:
Hurricanes can have dramatic short-term impacts on coastlines
Nor’easters and similar frequent winter storms are major contributors to coastline erosion over time

When water levels rise, large sediments (e.g. sand and gravel) are carried shoreward and deposited there. Over time, this process can cause entire coastlines to migrate inland.

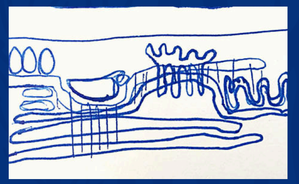


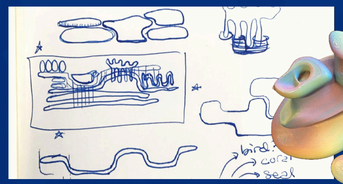
The migration of a typical barrier island in response to overwash or changing sea levels. Image: Julia Yee/Frontiers

TESTING

Feedback from american scouts aged from 11 to 15 years old

1. Improve Piece Fit – Ensure game pieces fit more securely together.
2. Simplify Rules – Make instructions clearer for quick, easy play.
3. Boost Interactivity – Add hands-on elements to show coastline changes dynamically.
4. Shorten Playtime – Consider faster rounds to keep players engaged.
5. Add Visuals – Use diagrams to explain erosion and rising sea levels.
6. Increase Strategy – Provide more decision points to engage players.
7. Include Fun Facts – Add cards with interesting coastal facts.
8. Team Option – Offer a team mode for more collaborative play.
9. Eco-Friendly Rewards – Reward choices that protect the environment.







NEXT STEPS

IInitially, we can make achievable improvements by refining the game pieces to align with the latest conceptual proposals, enlarging them, and ensuring a better fit. Additionally, we can expand the reflection cards and enhance the game’s learning curve. One of the key challenges, however, is securing funding for further design iterations. These should include bringing on a game specialist as a mentor, conducting additional testing with relevant communities, and providing further training in participatory design methods to improve the overall development process..

Riffe EC, Canfield M, Sabo AB, White AE and Zinnert JC (2024) What Are Barrier Islands and How Are They Affected by Climate Riffe EC, Canfield M, Sabo AB, White AE and Zinnert JC (2024) What Are Barrier Islands and How Are They Affected by Climate Change?. Front. Young Minds. 12:1347213. doi: 10.3389/frym.2024